MILITARY SPECIFICATION SHEET

HEAT SINKS, SEMICONDUCTOR DEVICES, CLASS 1 (FORMED)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the heat sinks described herein shall consist of this specification and the latest issue of MIL-H-87111.

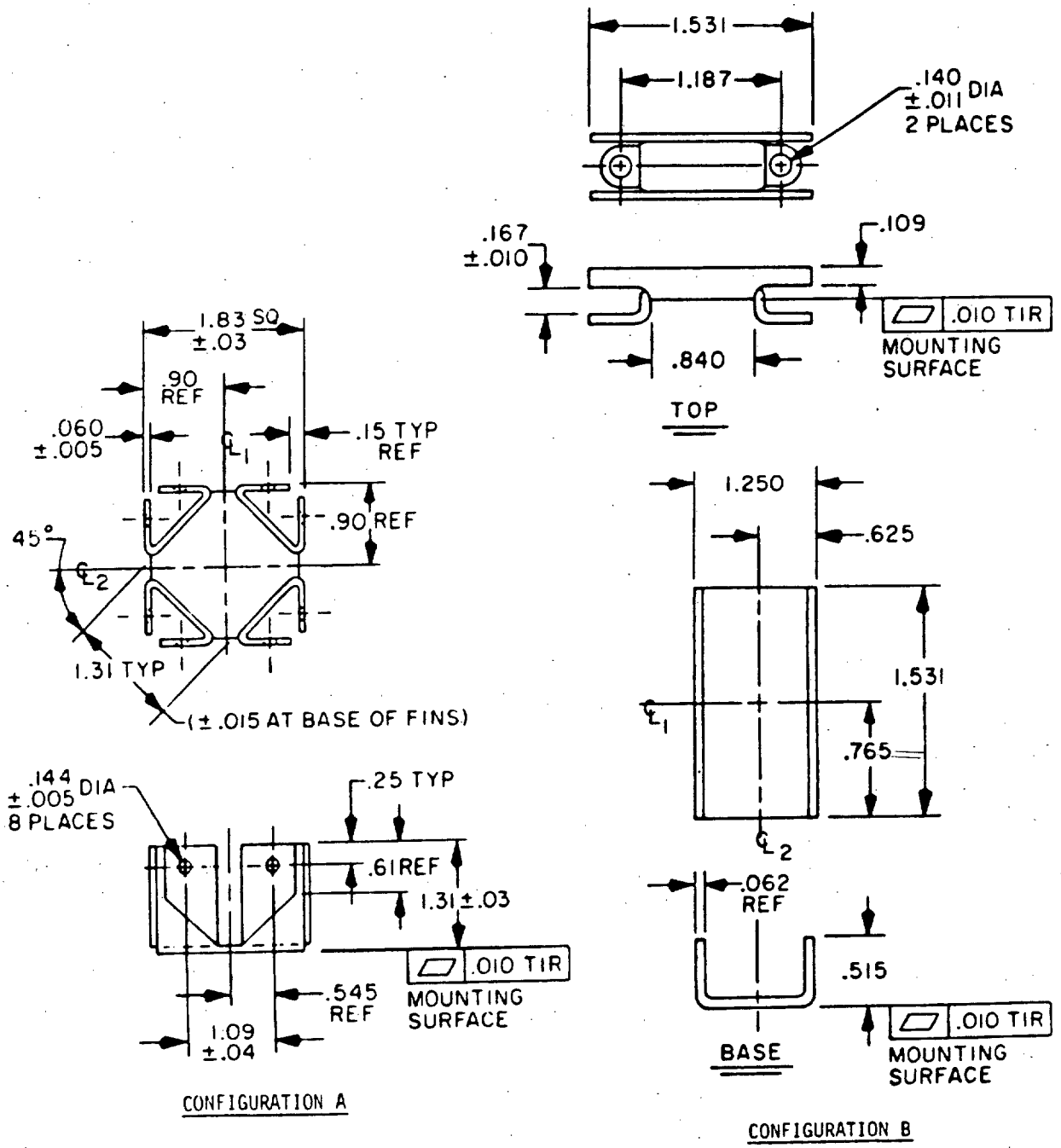
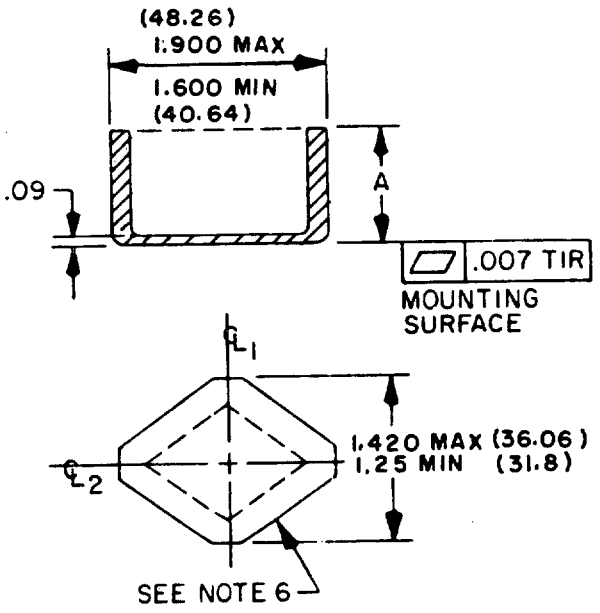


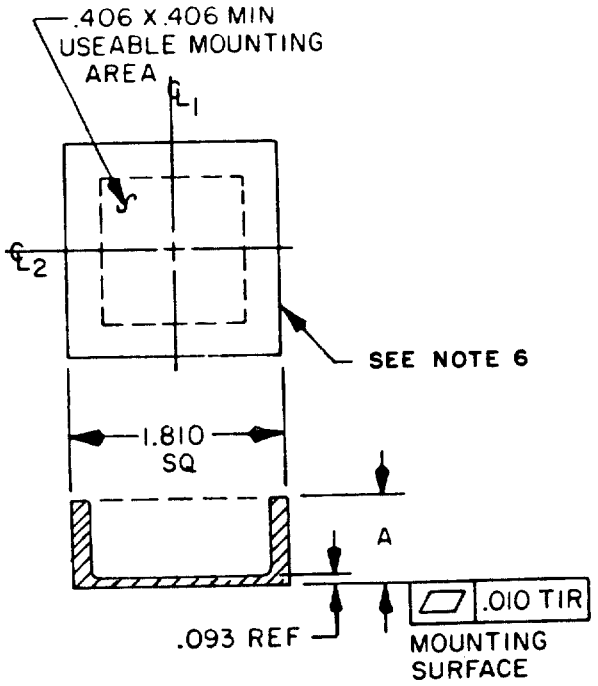
FIGURE 1. Heat sinks (formed).

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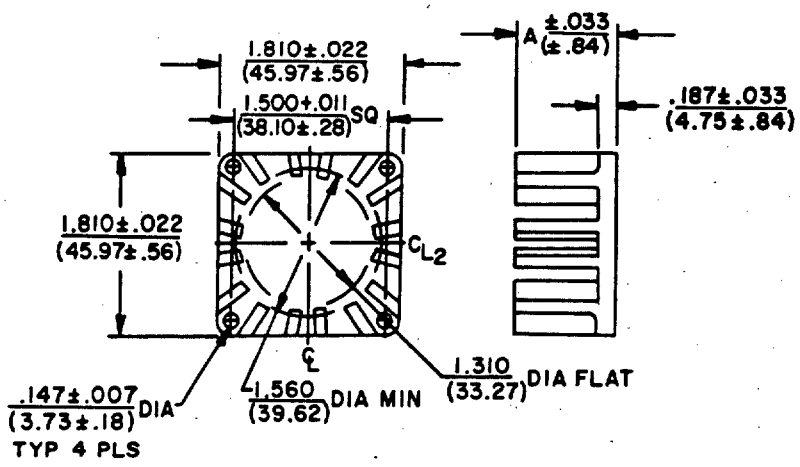
CONFIGURATION C



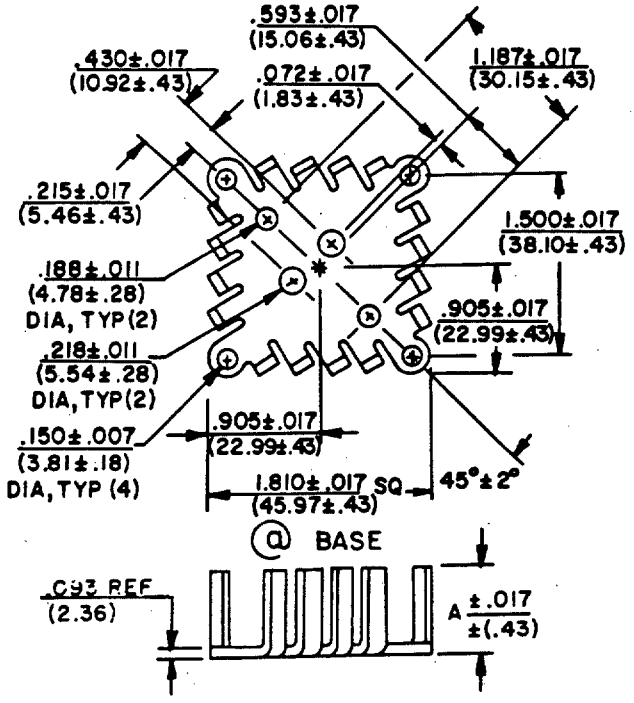
CONFIGURATION D (SEE NOTE 7)

FIGURE 1. Heat sinks (formed) - Continued.

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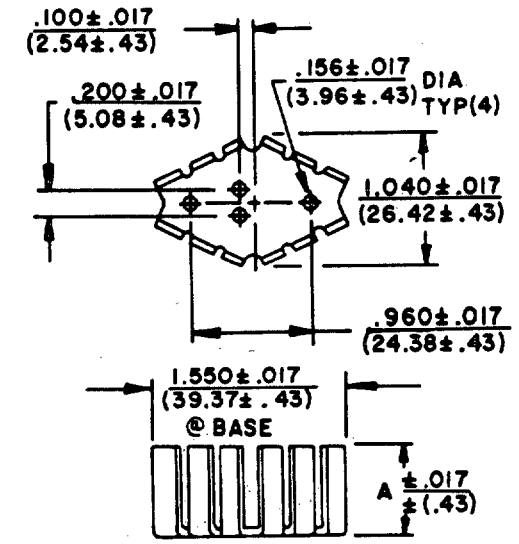
CONFIGURATION E



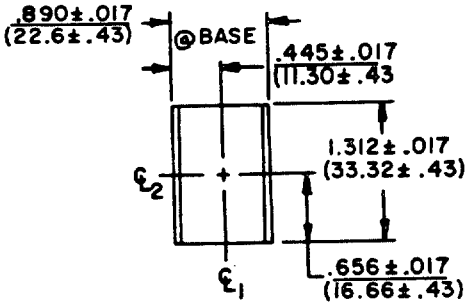
CONFIGURATION F

FIGURE 1. Heat sinks (formed) - Continued.

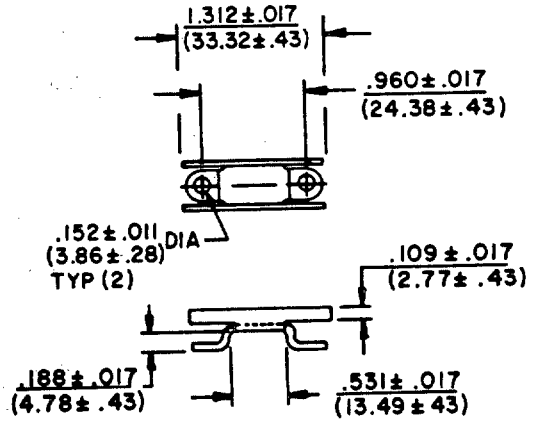
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CONFIGURATION G



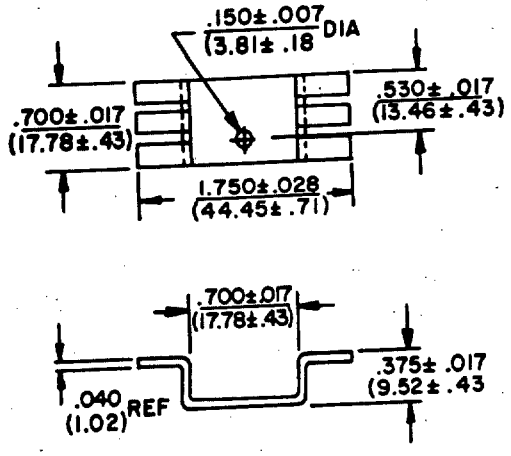
BASE
CONFIGURATION H



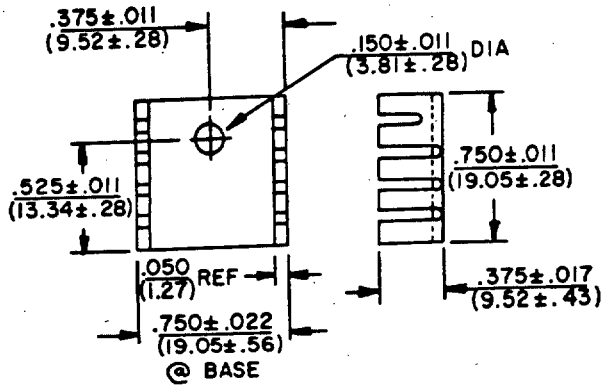
TOP

FIGURE 1. Heat sinks (formed) - Continued.

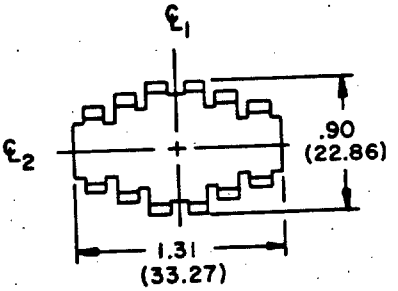
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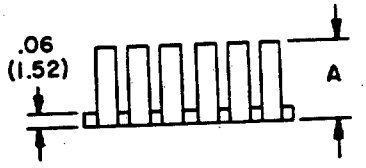


CONFIGURATION I



CONFIGURATION J





CONFIGURATION K

FIGURE 1. Heat sinks (formed) - Continued.

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1/ Part number M87111/05-																					
Hole pattern (see figure 2) 3/																			As shown in figure 1	Configuration	Dimension A (±0.25)	2/ Thermal resistance at 75°C rise
H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	H11	H12	H13	H14	H15	H16	H17	H18	H19				
1-001	1-011	1-021	1-031	1-041	1-051	1-061	1-071	1-081	1-091	1-101	1-111	1-121	1-131	1-141	1-151	1-161	1-171	1-181		A	---	4.2° C/W
1-002	1-012	1-022	1-032	1-042	1-052	1-062	1-072	1-082	1-092	1-102	1-112	1-122	1-132	1-142	1-152	1-162	1-172	1-182		B	---	11.0° C/W
1-003	1-013	1-023	1-033	1-043	1-053	1-063	1-073	1-083	1-093	1-103	1-113	1-123	1-133	1-143	1-153	1-163	1-173	1-183		C	.500 (12.70)	11.5° C/W
1-004	1-014	1-024	1-034	1-044	1-054	1-064	1-074	1-084	1-094	1-104	1-114	1-124	1-134	1-144	1-154	1-164	1-174	1-184		C	.750 (19.05)	8.6° C/W
1-005	1-015	1-025	1-035	1-045	1-055	1-065	1-075	1-085	1-095	1-105	1-115	1-125	1-135	1-145	1-155	1-165	1-175	1-185		C	1.000 (25.40)	6.9° C/W
1-006	1-016	1-026	1-036	1-046	1-056	1-066	1-076	1-086	1-096	1-106	1-116	1-126	1-136	1-146	1-156	1-166	1-176	1-186		C	1.250 (31.75)	6.2° C/W
1-007	1-017	1-027	1-037	1-047	1-057	1-067	1-077	1-087	1-097	1-107	1-117	1-127	1-137	1-147	1-157	1-167	1-177	1-187		D	.500 (12.70)	7.4° C/W
1-008	1-018	1-028	1-038	1-048	1-058	1-068	1-078	1-088	1-098	1-108	1-118	1-128	1-138	1-148	1-158	1-168	1-178	1-188		D	.750 (19.05)	5.9° C/W
1-009	1-019	1-029	1-039	1-049	1-059	1-069	1-079	1-089	1-099	1-109	1-119	1-129	1-139	1-149	1-159	1-169	1-179	1-189		D	1.000 (25.40)	4.4° C/W
1-010	1-020	1-030	1-040	1-050	1-060	1-070	1-080	1-090	1-100	1-110	1-120	1-130	1-140	1-150	1-160	1-170	1-180	1-190		D	1.250 (31.75)	3.3° C/W
1-191	1-200	1-209	1-218	1-227	1-236	1-245	1-254	1-263	1-272	1-281	1-290	1-299	1-308	1-317	1-326	1-335	1-344	1-353		E	.500 (12.70)	7.5° C/W
1-192	1-201	1-210	1-219	1-228	1-237	1-246	1-255	1-264	1-273	1-282	1-291	1-300	1-309	1-318	1-327	1-336	1-345	1-354		E	.750 (19.05)	6.2° C/W
1-193	1-202	1-211	1-220	1-229	1-238	1-247	1-256	1-265	1-274	1-283	1-292	1-301	1-310	1-319	1-328	1-337	1-346	1-355		E	1.000 (25.40)	5.0° C/W
1-194	1-203	1-212	1-221	1-230	1-239	1-248	1-257	1-266	1-275	1-284	1-293	1-302	1-311	1-320	1-329	1-338	1-347	1-356		E	1.250 (31.75)	4.2° C/W
																			1-362	F	.500 (12.70)	7.2° C/W
																			1-363	F	.750 (19.05)	6.1° C/W
																			1-364	F	1.000 (25.40)	5.2° C/W
																			1-365	F	1.250 (31.75)	4.2° C/W
																			1-366	G	.500 (12.70)	12.2° C/W
																			1-367	G	.750 (19.05)	7.9° C/W
																			1-368	G	1.000 (25.40)	6.7° C/W
1-195	1-204	1-213	1-222	1-231	1-240	1-249	1-258	1-267	1-276	1-285	1-294	1-303	1-312	1-321	1-330	1-339	1-348	1-357		H	---	14.5° C/W
																			1-369	I	---	16.0° C/W
																			1-370	J	---	21.0° C/W
1-196	1-205	1-214	1-223	1-232	1-241	1-250	1-259	1-268	1-277	1-286	1-295	1-304	1-313	1-322	1-331	1-340	1-349	1-358		K	.250 (6.35)	19.7° C/W
1-197	1-206	1-215	1-224	1-233	1-242	1-251	1-260	1-269	1-278	1-287	1-296	1-305	1-314	1-323	1-332	1-341	1-350	1-359		K	.500 (12.70)	19.7° C/W
1-198	1-207	1-216	1-225	1-234	1-243	1-252	1-261	1-270	1-279	1-288	1-297	1-306	1-315	1-324	1-333	1-342	1-351	1-360		K	.750 (19.05)	19.7° C/W
1-199	1-208	1-217	1-226	1-235	1-244	1-253	1-262	1-271	1-280	1-289	1-298	1-307	1-316	1-325	1-334	1-343	1-352	1-361		K	1.000 (25.40)	19.7° C/W

1/ Complete part number will include an additional letter to indicate finish (i.e., A, C, H or R). Example: M87111/05-1R010.
2/ Thermal resistance in natural convection.
3/ The hole pattern for dash numbers 1-362 through 1-368, 1-369, and 1-370 shall be as shown on figure 1.

FIGURE 1. Heat sinks (formed) - Continued.

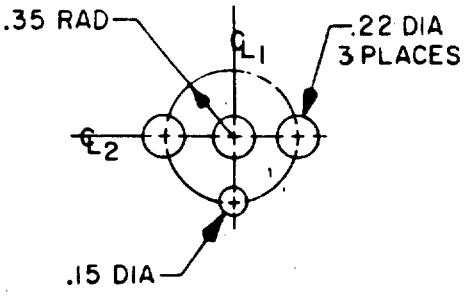
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INCHES	MM	INCHES	MM	INCHES	MM	INCHES	MM
.005	0.13	.09	2.3	.515	13.08	1.250	31.75
.007	0.18	.093	2.36	.545	13.84	1.31	33.3
.010	0.25	.109	2.77	.61	15.5	1.400	35.56
.011	0.28	.140	3.56	.625	15.88	1.531	38.89
.015	0.38	.144	3.66	.765	19.43	1.810	45.97
.03	0.8	.15	3.8	.840	21.34	1.83	46.5
.04	1.0	.167	4.24	.90	22.9	1.880	47.75
.060	1.52	.25	6.4	1.09	27.7		
.062	1.57	.406	10.31	1.187	30.15		

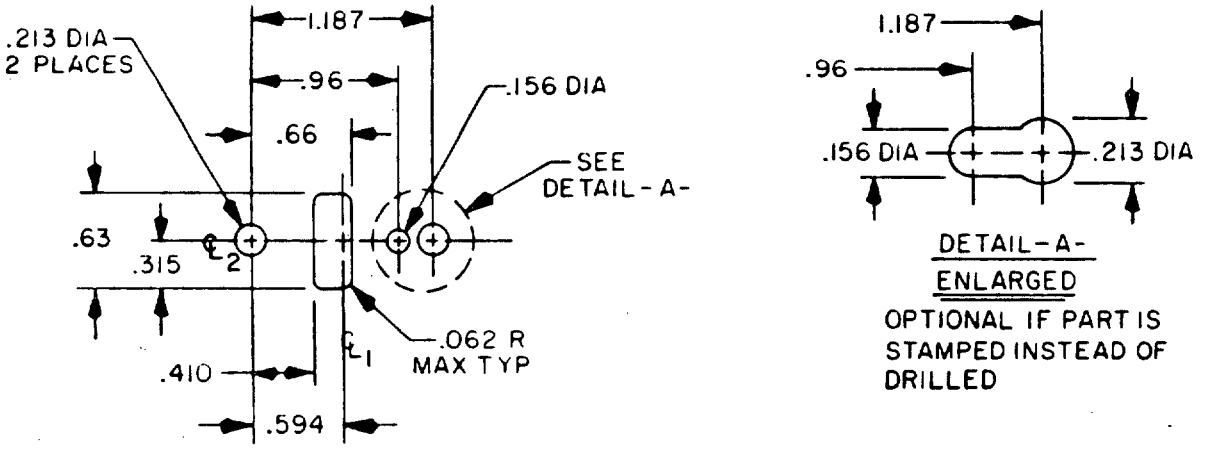
NOTES:
1. Dimensions are in inches.
2. Metric equivalents are in parentheses.
3. Metric equivalents are given for general information only.
4. Unless otherwise specified, tolerances are ±.017 (0.43 mm) on decimals and ±2° on angles. A .062 (1.57 mm) radius is permissible on corners and edges.
5. Drawing depicts only general design characteristics and is not intended to limit specified manufacturing processes.
6. Exterior fin design (staggered, in-line, etc.) optional within dimensions and thermal resistance specified.
7. Configuration D only, mounting hole patterns H2, H6 and H7 to be rotated 45° off centerline.

FIGURE 1. Heat sinks (formed) - Continued.

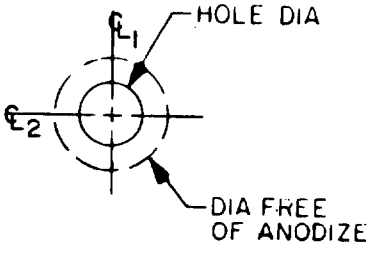
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H1 (TO-36 JEDEC CASE, TO-6 JEDEC CASE)



H2 (UNIVERSAL, TO-3, TO-66, TO-126 AND TO-220)

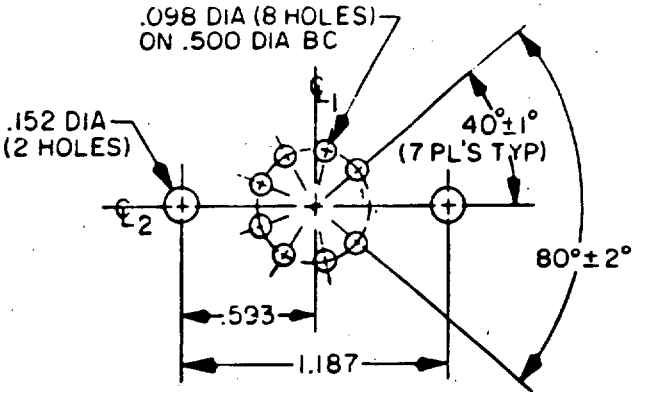


Hole Diameter	Diameter Free of Anodize ±.035(.89)	Hole pattern number
.200 ± .005 (5.08) (.13)	.500 (12.70)	H3
.270 ± .007 (6.86) (.18)	.750 (19.05)	H4
.330 ± .007 (8.38) (.18)	1.000 (25.40)	H5

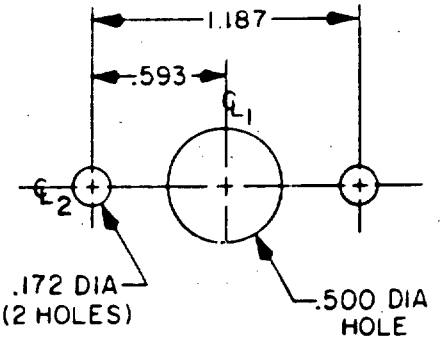
H3, H4 AND H5 (STUD)

FIGURE 2. Mounting hole patterns.

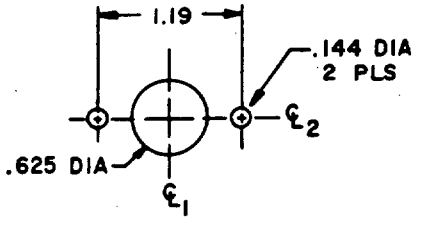
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H6 (TO-3 1C JEDEC case)



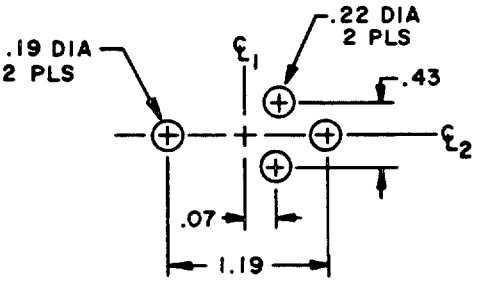
H7 (TO-8 1C JEDEC case)



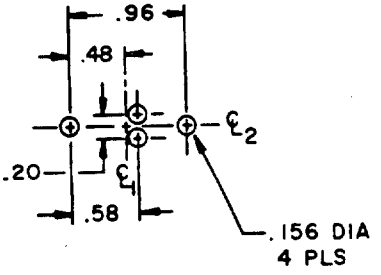
H8 (MULTI-LEAD TO-3)

FIGURE 2. Mounting hole patterns - Continued.

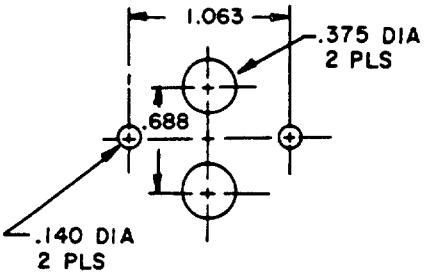
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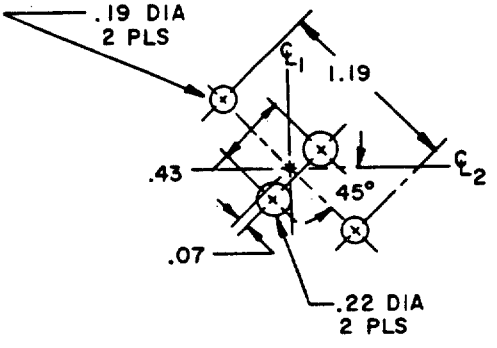
H9 (TO-3)



H10 (TO-66)



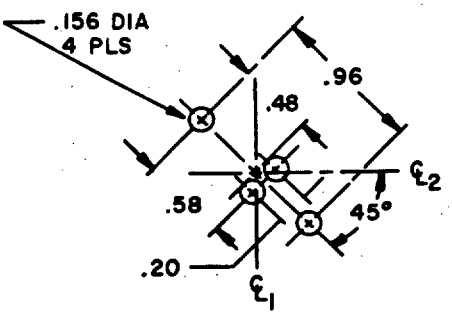
H11 (TWO TO-220's)



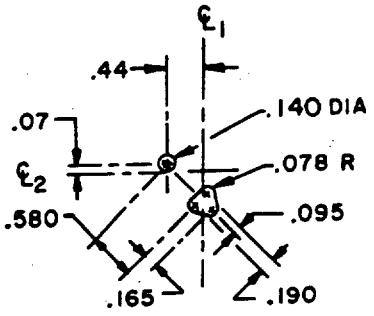
H12 (TO-3)

FIGURE 2. Mounting hole patterns - Continued.

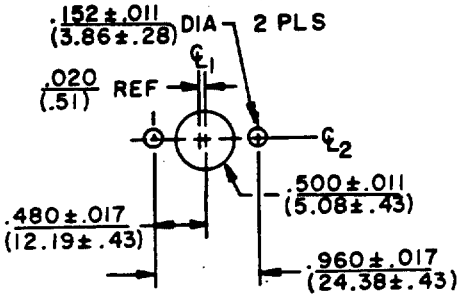
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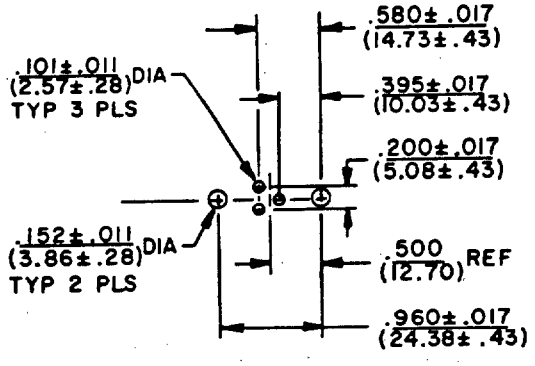
H13 (TO-66)



H14 (TO-220)



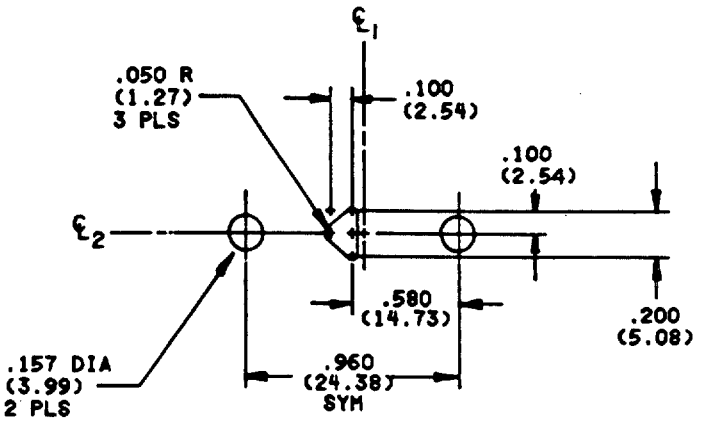
H15 (TO-66)



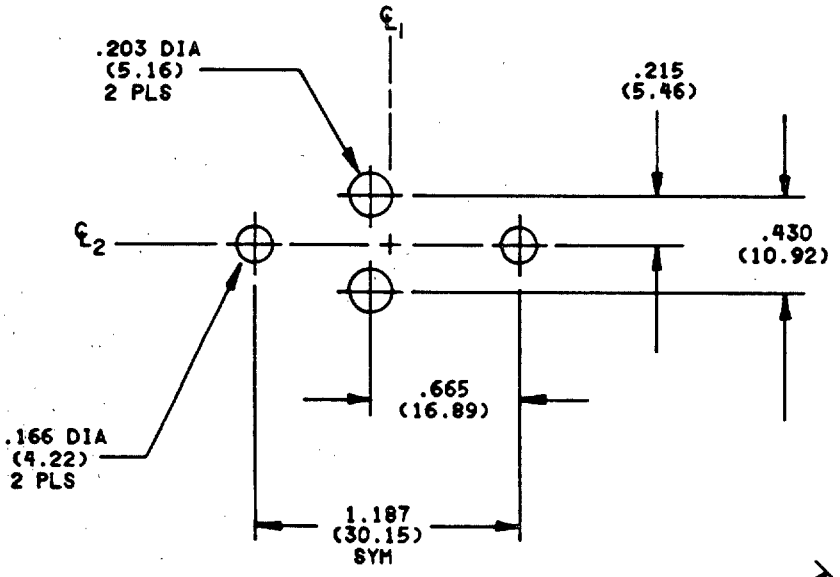
H16 (TO-66)

FIGURE 2. Mounting hole patterns - Continued.

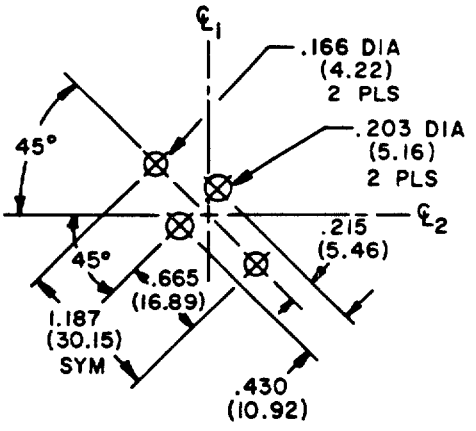
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H17 (TO-66)



H18 (TO-3)



H19 (TO-3)

FIGURE 2. Mounting hole patterns - Continued.

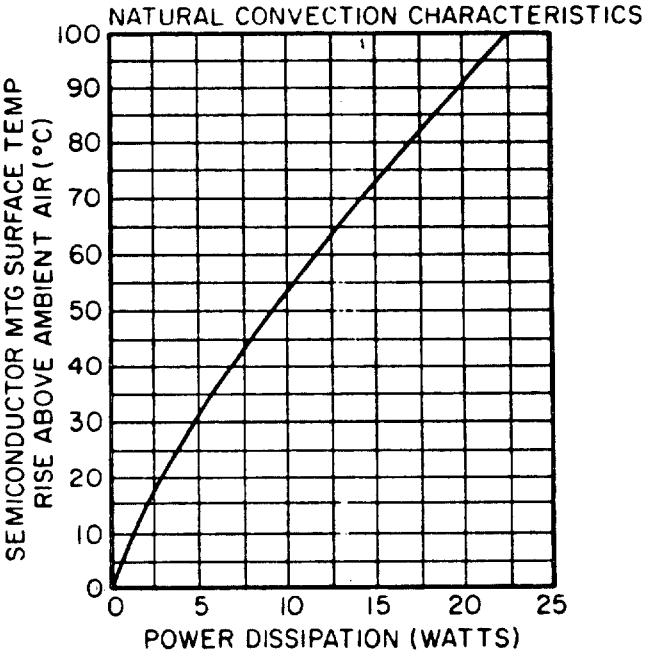
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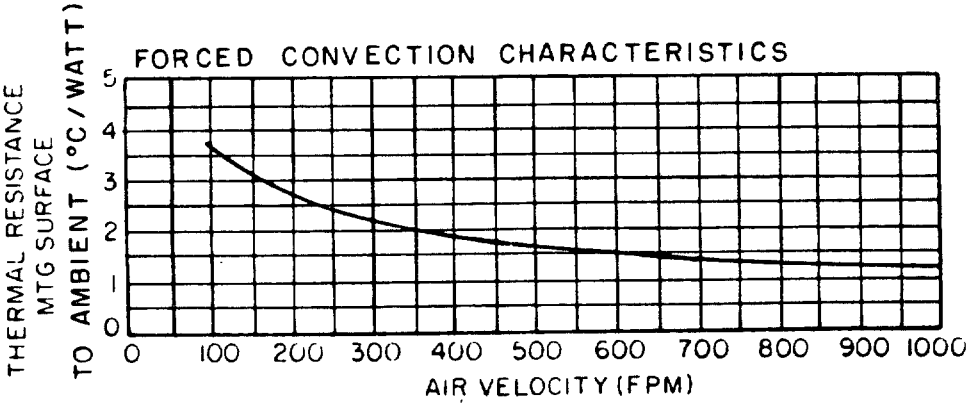
INCHES	MM	INCHES	MM	INCHES	MM	INCHES	MM	INCHES	MM
.005	0.13	.144	3.66	.200	5.08	.43	10.92	.63	16.0
.007	0.18	.15	3.8	.213	5.41	.44	11.18	.66	16.8
.035	0.89	.152	3.86	.22	5.6	.48	12.2	.688	17.5
.062	1.57	.156	3.96	.270	6.86	.500	12.7	.750	19.1
.07	1.78	.165	4.19	.315	8.00	.58	14.7	.96	24.4
.078	1.98	.172	4.37	.330	8.38	.580	14.73	1.000	25.40
.095	2.41	.19	4.8	.35	8.9	.593	15.06	1.063	27.00
.098	2.49	.190	4.82	.375	9.53	.594	15.09	1.187	30.15
.140	3.56	.20	5.1	.410	10.41	.625	15.88	1.19	30.2

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Metric equivalents are in parentheses.
4. Unless otherwise specified, tolerance is ±.010 (0.25 mm) on decimals and ±2° on angles. A .062 (1.57 mm) radius is permissible on corners.
5. Drawing depicts only general design characteristics and is not intended to limit specific manufacturing processes.

FIGURE 2. Mounting hole patterns - Continued.

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CONFIGURATION A

FIGURE 3. Thermal performance characteristic.

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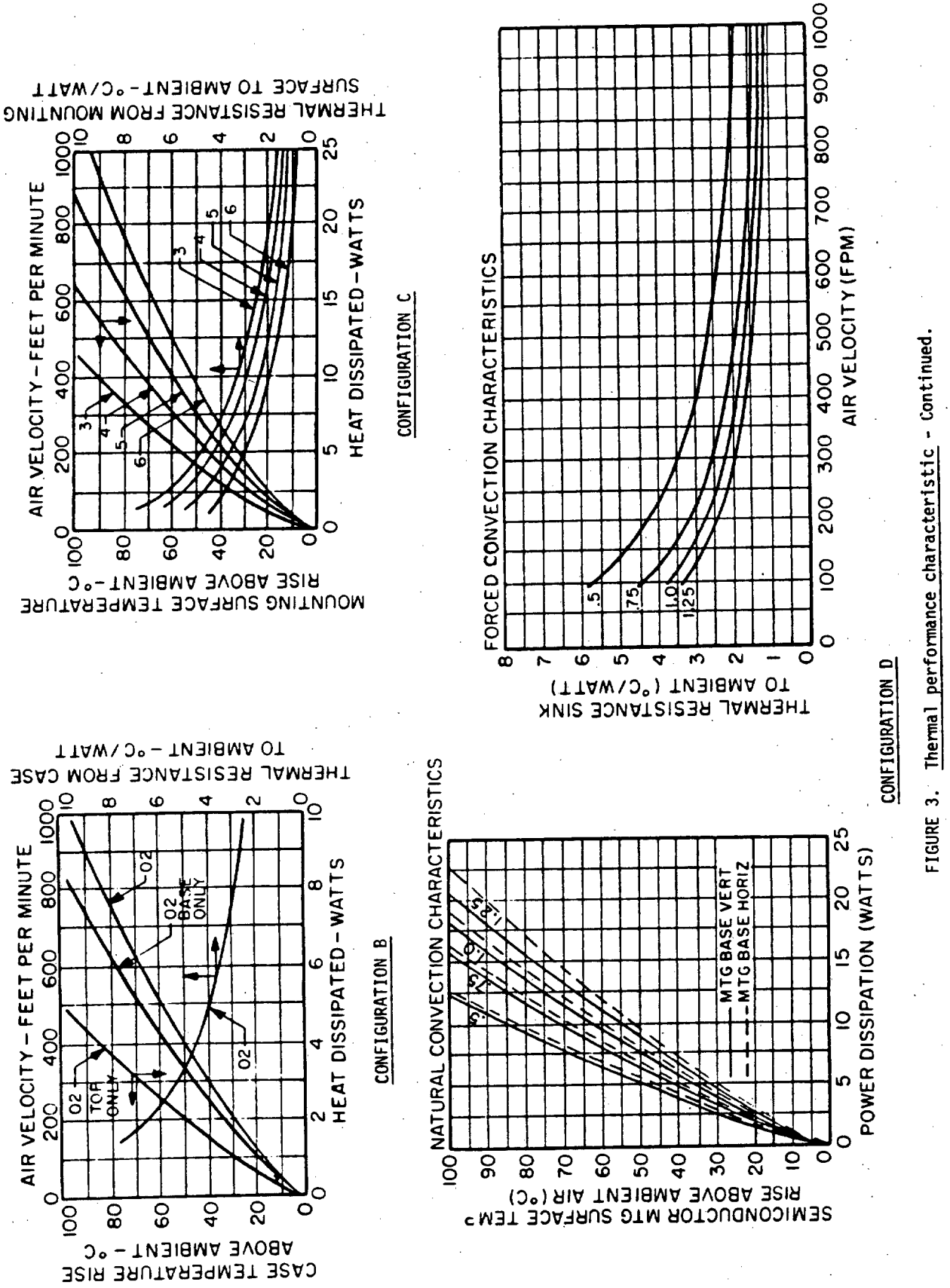
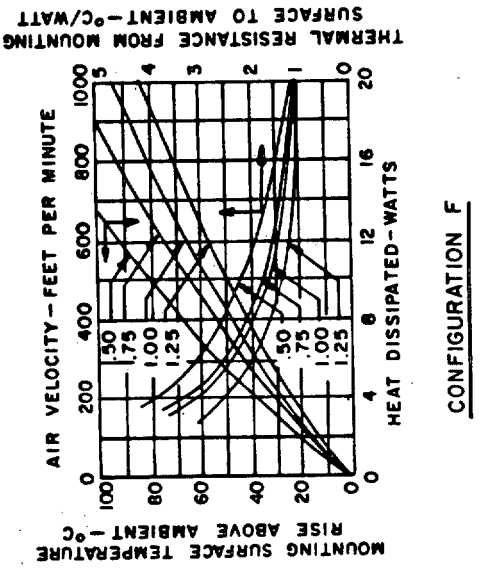
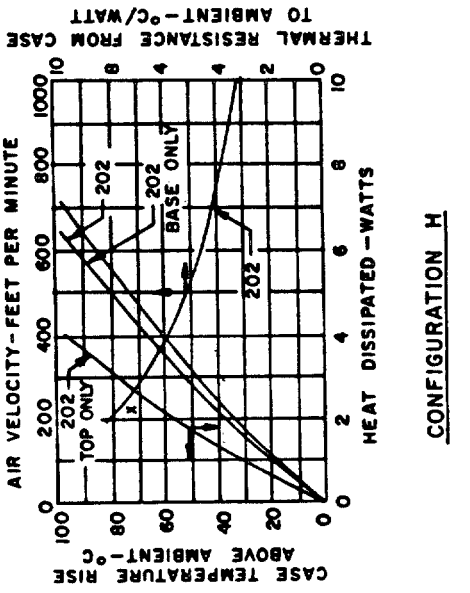


FIGURE 3. Thermal performance characteristic - Continued.

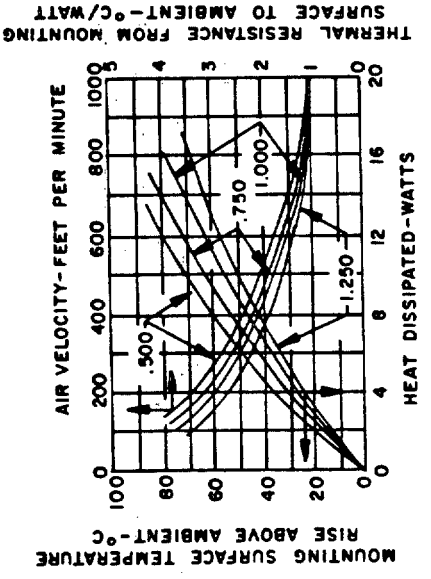
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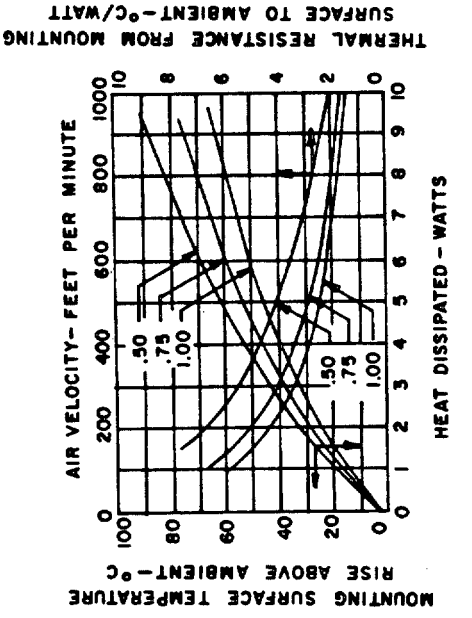
CONFIGURATION F



CONFIGURATION H



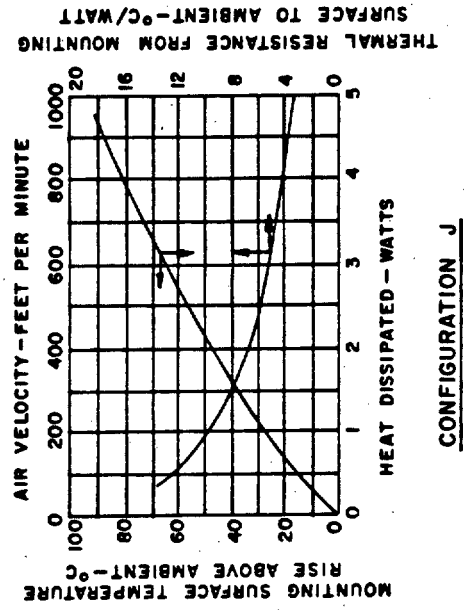
CONFIGURATION E



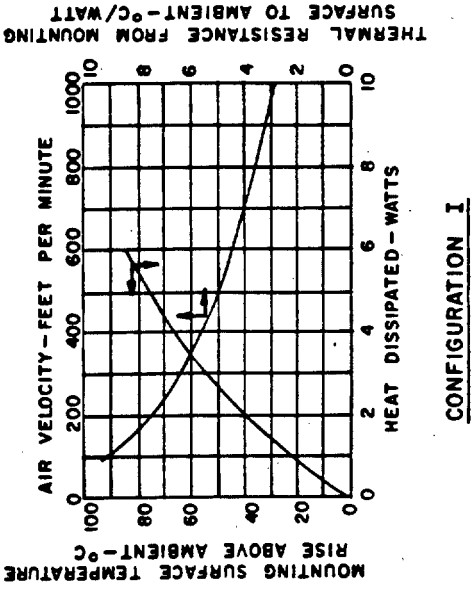
CONFIGURATION G

FIGURE 3. Thermal performance characteristic - Continued.

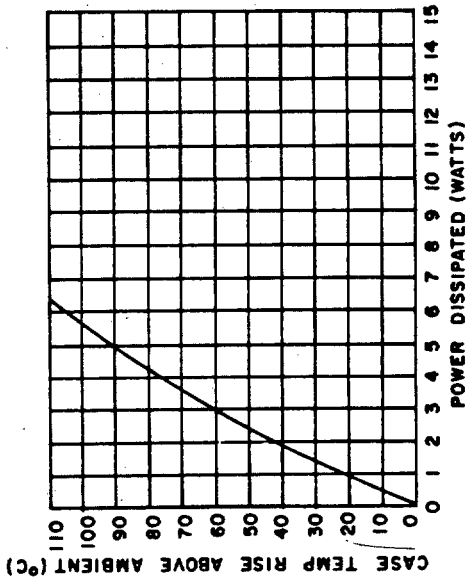
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CONFIGURATION J



CONFIGURATION I



CONFIGURATION K

FIGURE 3. Thermal performance characteristic - Continued.

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REQUIREMENTS:

Design and configuration: See figures 1 and 2.

Material: Aluminum alloy (class 1).

Finish: Black anodize, insulating coating, hard anodize, or gold chromate (finish A, C, H, or R).

Mounting surface flatness: See figure 1.

Surface roughness: Not applicable.

Thermal resistance: See figure 1.

JEDEC case size: See figure 2.

Part number: M87111/05- (dash number from figure 1).

NOTE: The recommended thermal performance characteristics in forced air and natural convection, as opposed to power dissipation, is shown on figure 3.

Revision letters are not used to denote changes due to the extensiveness of the changes.

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Navy - TD
Air Force - 11, 19, 99
DLA - ES

User activities:
Navy - AS, CG, MC, SH

Preparing activity:
Air Force - 85

Agent:
DLA - ES

(Project 5999-0154-3)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-H-87111/5B	HEAT SINKS, SEMICONDUCTOR DEVICE, CLASS 2 AND 3 (RETAINWR CLIP TYPE)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

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PREVIOUS EDITION IS OBSOLETE.

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